Modelling a 3D printer

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0 Introduction

3D Printers have the basic task of extruding filament at a given x, y, z coordinate at time t. The mechanism that achieves that movement determines the machine's costs, print quality, and robustness. Out of many different options, the Core XY mechanism is a popular choice for modern printers. Different than Cartesian designs, where one motor drives one axis, here each motor simultaneously changes the x and y position of the tool head (see Figure 1). This results in a more complex kinematic model, but allows a more sturdy casing and faster printing speeds. Our task is to model and simulate the system.

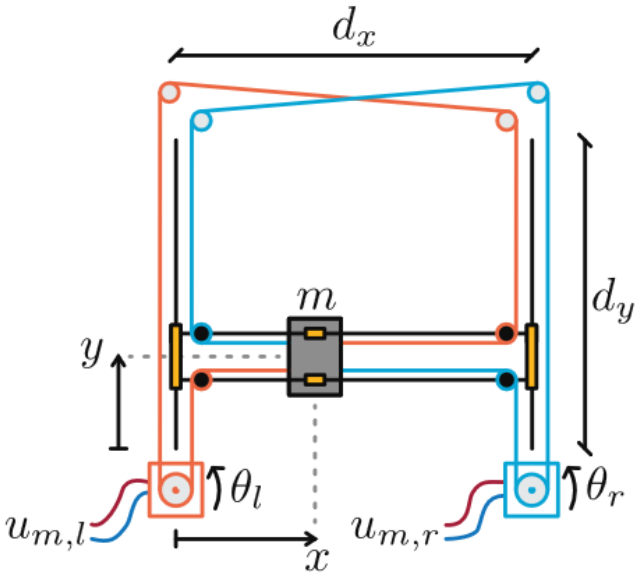


Figure 1: Schematic of the Core XY mechanism.

0.1 Symbol table

Table 1: Symbols and parameter values.

Description	Symbol	Value
Motor angle left	θ_L	–
Motor voltage left	$u_{m,L}$	–
Motor current left	i_L	–
Motor radius	r_m	11 mm
Motor induction	L_m	10 mH
Motor resistor	R_m	90 Ω
Motor inertia	J_m	8×10^{-6} kg m^2
Motor constant	K_m	0.07 N m/A
Tool head position	x, y	–
Tool head mass	m	60 g
Nozzle voltage	u_n	–
Nozzle current	i_n	–
Nozzle capacitor	C_n	–
Nozzle thermal capacity	$C_{t,n}$	–
Nozzle thermal resistance	$R_{t,n}$	9 K/W
Nozzle resistor	R_n	8 Ω
Nozzle temperature	T_n	240 °C
Environmental temperature	T_e	300 K
Bearing friction rolling	μ_r	2.7×10^{-8} N m s/rad
Bearing friction linear x-direction	$\mu_{l,x}$	0.002 N s/m
Bearing friction linear y-direction	$\mu_{l,y}$	0.001 N s/m
Bearing radius	r_b	3 mm
Print bed temperature	T_b	105 °C
Print area dimension x	d_x	29 cm
Print area dimension y	d_y	29 cm

1 Physical Modelling

1.1 Kinematics

First task is to derive the direct kinematics of the Core XY mechanism. One fact that slightly complicates this is the fact that rotating one motor will move the tool head in both x and y directions. By removing the right motor, we can see that by moving the tool head in the positive x direction **or** in the positive y direction, the left motor will rotate in the positive direction. We can draw similar conclusions for the right motor, for which we get the following equations:

$$\begin{aligned} \Delta L &= \Delta x + \Delta y \\ \Delta R &= \Delta x - \Delta y \\ \Delta x &= \frac{\Delta L + \Delta R}{2} \\ \Delta y &= \frac{\Delta L - \Delta R}{2} \end{aligned} \tag{1.1}$$

With ΔL and ΔR being the length of belt moved by the motors, and Δx and Δy being the change in position of the tool head.

$$\begin{aligned} \Delta L &= r_m \, \theta_L \\ \Delta R &= r_m \, \theta_R \end{aligned} \tag{1.2}$$

If we assume that in the beginning the motors are at $\theta_L = \theta_R = 0$ and the cart is at $(0, 0)$, we get the following equations:

$$\begin{aligned} x(\theta_L, \theta_R) &= \frac{r_m \theta_L + r_m \theta_R}{2} = \frac{r_m}{2} \left(\theta_L + \theta_R\right) \\ y(\theta_L, \theta_R) &= \frac{r_m \theta_L - r_m \theta_R}{2} = \frac{r_m}{2} \left(\theta_L - \theta_R\right) \end{aligned} \tag{1.3}$$

$$\begin{aligned} \dot{x}(\dot{\theta}_L, \dot{\theta}_R) &= \frac{r_m}{2} \left(\dot{\theta}_L + \dot{\theta}_R\right) \\ \dot{y}(\dot{\theta}_L, \dot{\theta}_R) &= \frac{r_m}{2} \left(\dot{\theta}_L - \dot{\theta}_R\right) \end{aligned} \tag{1.4}$$

1.2 Dynamics

In the next part we model the dynamics of the 3D printer system by applying the Euler-Lagrange equation. Additionally, we will split the problem to a mechanical and an electrical one, with the angular velocity of the motor shafts connecting them.

Because of that, we choose generalized coordinates:

$$q = \begin{bmatrix} x \\ y \\ q_L \\ q_R \end{bmatrix} \qquad \dot{q} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ i_L \\ i_R \end{bmatrix} \tag{1.5}$$

We can also then split them among our two problems:

$$\dot{q}_m = \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} \text{ (mechanical part)}, \qquad \dot{q}_e = \begin{bmatrix} i_L \\ i_R \end{bmatrix} \text{ (electrical part)} \tag{1.6}$$

1.2.1 Mechanical part

Kinetic energy

Firstly, we derive the kinetic energy terms. In the *mechanical* part, the only mass is the tool head. With that we have:

$$K_{mech} = \tfrac{1}{2} m(\dot{x}^2 + \dot{y}^2) \tag{1.7}$$

As the whole system is planar, we can see that there is no potential energy terms as it will be constant.

Dissipative energy

Secondly, we find the dissipative energy terms. Each motor has 4 rotational bearings, and 4 linear bearings for the whole system, two along each axis. We define r_b as the bearing radius, later it will be kept the same as the motor radius for simplicity. From (1.3), we can get transformations (1.8) with trivial time derivatives.

$$\theta_{b,L} = \frac{x+y}{r_b}, \qquad \theta_{b,R} = \frac{x-y}{r_b} \tag{1.8}$$

Then, for each side we get rotational dissipative energy:

$$D_L = 4 \cdot \tfrac{1}{2} \cdot \dot{\theta}_{b,L}^2 \, \mu_r = 2\mu_r \frac{1}{r_b^2} \left[\dot{x}^2 + 2\dot{x}\dot{y} + \dot{y}^2\right]$$

$$D_R = 4 \cdot \tfrac{1}{2} \cdot \dot{\theta}_{b,R}^2 \, \mu_r = 2\mu_r \frac{1}{r_b^2} \left[\dot{x}^2 - 2\dot{x}\dot{y} + \dot{y}^2\right]$$

Then simply combining them:

$$\boxed{D_{\text{rot}} = 4 \, \frac{\mu_r}{r_b^2} \left[\dot{x}^2 + \dot{y}^2\right]}$$

Similarly, we get the linear dissipative energy:

$$D_{\text{lin}} = 2 \cdot \tfrac{1}{2} \mu_{l,x} \, \dot{x}^2 + 2 \cdot \tfrac{1}{2} \mu_{l,y} \, \dot{y}^2 = \mu_{l,x} \, \dot{x}^2 + \mu_{l,y} \, \dot{y}^2$$

Then, combining the terms we get:

$$D_m = \left[4 \, \frac{\mu_r}{r_b^2} + \mu_{l,x}\right] \dot{x}^2 + \left[4 \, \frac{\mu_r}{r_b^2} + \mu_{l,y}\right] \dot{y}^2 \tag{1.9}$$

1.2.2 Electromechanical part

Now we model the energies of the electrical part of the system. Keep in mind that both motors are the same. Below we can see the equivalent circuit of a DC motor:

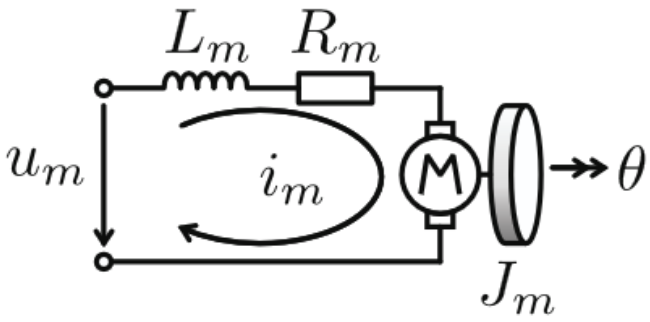


Figure 2: Equivalent circuit of a DC motor.

Kinetic energy

The inductor and the inertial element will contribute to the kinetic energy of this part of the system. Similarly, like (1.8), using the motor shaft radius:

$$K_L = \tfrac{1}{2} i_L^2 L_m + \tfrac{1}{2} J_m \dot{\theta}_L^2 = \tfrac{1}{2} i_L^2 L_m + \tfrac{1}{2} J_m \frac{1}{r_m^2} \left(\dot{x}^2 + \dot{y}^2 + 2\dot{x}\dot{y}\right)$$
$$K_R = \tfrac{1}{2} i_R^2 L_m + \tfrac{1}{2} J_m \frac{1}{r_m^2} \left(\dot{x}^2 + \dot{y}^2 - 2\dot{x}\dot{y}\right)$$

Combining them we get:

$$K_{EL} = \tfrac{1}{2} L_m \left[i_R^2 + i_L^2\right] + J_m \frac{1}{r_m^2} \left(\dot{x}^2 + \dot{y}^2\right) \tag{1.10}$$

As there are no capacitors in the system, there is no potential energy change as in the mechanical part.

Dissipative energy

Only dissipative term in the electrical part of the system is the resistor:

$$\begin{gathered} D_{EL,L} = \tfrac{1}{2} i_L^2 R_m, \qquad D_{EL,R} = \tfrac{1}{2} i_R^2 R_m \\ D_{EL} = \tfrac{1}{2} R_m \left(i_L^2 + i_R^2\right) \end{gathered} \tag{1.11}$$

Matrix equations

Finally, we realize that the system is linear, and write the matrix expressions:

$$K = \tfrac{1}{2}\, \dot{q}^\top \mathbf{M}\, \dot{q}, \qquad D = \tfrac{1}{2}\, \dot{q}^\top \mathbf{N}\, \dot{q} \tag{1.12}$$

With corresponding matrices:

$$\mathbf{M} = \begin{bmatrix} m + J_m \dfrac{2}{r_m^2} & 0 & 0 & 0 \\ 0 & m + J_m \dfrac{2}{r_m^2} & 0 & 0 \\ 0 & 0 & L_m & 0 \\ 0 & 0 & 0 & L_m \end{bmatrix}$$
$$\mathbf{N} = \begin{bmatrix} 8\dfrac{\mu_r}{r_b^2} + 2\mu_{l,x} & 0 & 0 & 0 \\ 0 & 8\dfrac{\mu_r}{r_b^2} + 2\mu_{l,y} & 0 & 0 \\ 0 & 0 & R_m & 0 \\ 0 & 0 & 0 & R_m \end{bmatrix} \tag{1.13}$$

1.2.3 Generalized force

Mechanical coordinates

The non-conservative forces in the mechanical part are given by the motor torques τ_L and τ_R. The virtual work they perform is:

$$\Delta W = \tau_L \, \Delta\theta_L + \tau_R \, \Delta\theta_R$$

With the familiar transformation, analogous to (1.8) but using the motor radius r_m, we get:

$$\Delta W = \frac{\tau_L + \tau_R}{r_m} \, \Delta x + \frac{\tau_L - \tau_R}{r_m} \, \Delta y$$

Applying the motor law $\tau = K_m i$, the generalized forces for x and y are:

$$Q_x = \frac{K_m}{r_m} \left(i_L + i_R \right), \qquad Q_y = \frac{K_m}{r_m} \left(i_L - i_R \right) \tag{1.14}$$

Electrical coordinates

The non-conservative forces in the electrical part are the input voltages and the back-EMF of the motors:

$$\Delta W = \left(u_{m,L} - K_m \dot{\theta}_L \right) \Delta q_L + \left(u_{m,R} - K_m \dot{\theta}_R \right) \Delta q_R$$

Using the same transformation, we get the generalized forces for q_L and q_R:

$$Q_{q_L} = u_{m,L} - \frac{K_m}{r_m} \left(\dot{x} + \dot{y} \right), \qquad Q_{q_R} = u_{m,R} - \frac{K_m}{r_m} \left(\dot{x} - \dot{y} \right) \tag{1.15}$$

Lagrange equations

Using the fact this is a linear system and there are no potential energies we get:

$$\mathcal{L} = K - V = \tfrac{1}{2} \, \dot{q}^\top \mathbf{M} \, \dot{q} - V, \qquad V = 0 \quad \Rightarrow \quad \mathcal{L} = \tfrac{1}{2} \, \dot{q}^\top \mathbf{M} \, \dot{q}$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}} = \frac{d}{dt} \left(\mathbf{M} \dot{q} \right) = \mathbf{M} \ddot{q}$$
$$\frac{\partial \mathcal{L}}{\partial q} = 0$$
$$\frac{\partial D}{\partial \dot{q}} = \mathbf{N} \dot{q}$$

And finally solving the Euler-Lagrange equation:

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}} - \frac{\partial \mathcal{L}}{\partial q} = -\frac{\partial D}{\partial \dot{q}} + \mathbf{Q}, \qquad \mathbf{Q} = \begin{bmatrix} Q_x \\ Q_y \\ Q_{q_L} \\ Q_{q_R} \end{bmatrix}$$

$$\mathbf{M}\ddot{q} + \mathbf{N}\dot{q} = \mathbf{Q} \tag{1.16}$$

Expanding back out of matrix form we get:

$$\begin{aligned}
\left(m + J_m\frac{2}{r_m^2}\right)\ddot{x} + \left(8\frac{\mu_r}{r_b^2} + 2\mu_{l,x}\right)\dot{x} - \frac{K_m}{r_m}i_L - \frac{K_m}{r_m}i_R = 0 \\
\left(m + J_m\frac{2}{r_m^2}\right)\ddot{y} + \left(8\frac{\mu_r}{r_b^2} + 2\mu_{l,y}\right)\dot{y} - \frac{K_m}{r_m}i_L + \frac{K_m}{r_m}i_R = 0 \\
L_m\frac{di_L}{dt} + R_m i_L + \frac{K_m}{r_m}\dot{x} + \frac{K_m}{r_m}\dot{y} - u_{m,L} = 0 \\
L_m\frac{di_R}{dt} + R_m i_R + \frac{K_m}{r_m}\dot{x} - \frac{K_m}{r_m}\dot{y} - u_{m,R} = 0
\end{aligned}$$

Or with substitutions, we get final form of the four differential equations:

$$m_{eq} = m + \frac{2J_m}{r_m^2}, \qquad b_x = 8\frac{\mu_r}{r_b^2} + 2\mu_{l,x}, \qquad b_y = 8\frac{\mu_r}{r_b^2} + 2\mu_{l,y}$$

$$\boxed{\begin{aligned}
\ddot{x} &= -\frac{b_x}{m_{eq}}\dot{x} + \frac{K_m}{m_{eq}r_m}\left(i_L + i_R\right) \\
\ddot{y} &= -\frac{b_y}{m_{eq}}\dot{y} + \frac{K_m}{m_{eq}r_m}\left(i_L - i_R\right) \\
\frac{di_L}{dt} &= -\frac{K_m}{r_m L_m}\left(\dot{x} + \dot{y}\right) - \frac{R_m}{L_m}i_L + \frac{u_{m,L}}{L_m} \\
\frac{di_R}{dt} &= -\frac{K_m}{r_m L_m}\left(\dot{x} - \dot{y}\right) - \frac{R_m}{L_m}i_R + \frac{u_{m,R}}{L_m}
\end{aligned}} \tag{1.17}$$

Or in **state-space** form:

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \dot{x} \\ \dot{y} \\ i_L \\ i_R \end{bmatrix}, \qquad \dot{\mathbf{x}} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \ddot{x} \\ \ddot{y} \\ \dfrac{di_L}{dt} \\ \dfrac{di_R}{dt} \end{bmatrix}, \qquad \mathbf{u} = \begin{bmatrix} u_{m,L} \\ u_{m,R} \end{bmatrix}$$

$$\dot{\mathbf{x}} = \mathbf{A}\,\mathbf{x} + \mathbf{B}\,\mathbf{u} \tag{1.18}$$

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{b_x}{m_{eq}} & 0 & \frac{K_m}{m_{eq} r_m} & \frac{K_m}{m_{eq} r_m} \\ 0 & 0 & 0 & -\frac{b_y}{m_{eq}} & \frac{K_m}{m_{eq} r_m} & -\frac{K_m}{m_{eq} r_m} \\ 0 & 0 & -\frac{K_m}{r_m L_m} & -\frac{K_m}{r_m L_m} & -\frac{R_m}{L_m} & 0 \\ 0 & 0 & -\frac{K_m}{r_m L_m} & \frac{K_m}{r_m L_m} & 0 & -\frac{R_m}{L_m} \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{L_m} & 0 \\ 0 & \frac{1}{L_m} \end{bmatrix}$$

1.2.4 Decoupling the x and y dynamics

To decouple the system, we introduce a set of useful substitutions:

$$\boxed{\begin{aligned} i_x &= i_L + i_R, & u_{m,x} &= u_{m,L} + u_{m,R} \\ i_y &= i_L - i_R, & u_{m,y} &= u_{m,L} - u_{m,R} \\ i_L &= \frac{i_x + i_y}{2}, & u_{m,L} &= \frac{u_{m,x} + u_{m,y}}{2} \\ i_R &= \frac{i_x - i_y}{2}, & u_{m,R} &= \frac{u_{m,x} - u_{m,y}}{2} \end{aligned}} \tag{1.19}$$

By substituting our new decoupled currents we get:

$$\begin{aligned} \frac{di_L}{dt} &= \tfrac{1}{2}\frac{di_x}{dt} + \tfrac{1}{2}\frac{di_y}{dt} = -\frac{K_m}{r_m L_m}\dot{x} - \frac{K_m}{r_m L_m}\dot{y} - \frac{R_m}{2L_m} i_x - \frac{R_m}{2L_m} i_y + \frac{1}{2L_m}\left(u_{m,x} + u_{m,y}\right) \\ \frac{di_R}{dt} &= \tfrac{1}{2}\frac{di_x}{dt} - \tfrac{1}{2}\frac{di_y}{dt} = -\frac{K_m}{r_m L_m}\dot{x} + \frac{K_m}{r_m L_m}\dot{y} - \frac{R_m}{2L_m} i_x + \frac{R_m}{2L_m} i_y + \frac{1}{2L_m}\left(u_{m,x} - u_{m,y}\right) \end{aligned}$$

Then by adding and subtracting the two equations to cancel terms we get:

$$\begin{aligned} \frac{di_x}{dt} &= -\frac{2K_m}{r_m L_m}\dot{x} - \frac{R_m}{L_m} i_x + \frac{1}{L_m} u_{m,x} \\ \frac{di_y}{dt} &= -\frac{2K_m}{r_m L_m}\dot{y} - \frac{R_m}{L_m} i_y + \frac{1}{L_m} u_{m,y} \end{aligned} \tag{1.20}$$

And by substituting (1.19) into the first two equations of (1.17):

$$\begin{aligned} \ddot{x} &= -\frac{b_x}{m_{eq}}\dot{x} + \frac{K_m}{m_{eq} r_m} i_x \\ \ddot{y} &= -\frac{b_y}{m_{eq}}\dot{y} + \frac{K_m}{m_{eq} r_m} i_y \end{aligned} \tag{1.21}$$

And finally:

$$\tilde{\mathbf{x}} = \begin{bmatrix} x \\ \dot{x} \\ i_x \\ y \\ \dot{y} \\ i_y \end{bmatrix}, \qquad \dot{\tilde{\mathbf{x}}} = \begin{bmatrix} \dot{x} \\ \ddot{x} \\ \dfrac{di_x}{dt} \\ \dot{y} \\ \ddot{y} \\ \dfrac{di_y}{dt} \end{bmatrix}, \qquad \tilde{\mathbf{u}} = \begin{bmatrix} u_{m,x} \\ u_{m,y} \end{bmatrix}$$

$$\dot{\tilde{\mathbf{x}}} = \tilde{\mathbf{A}}\,\tilde{\mathbf{x}} + \tilde{\mathbf{B}}\,\tilde{\mathbf{u}} \tag{1.22}$$

$$\tilde{\mathbf{A}} = \left[\begin{array}{ccc|ccc} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & -\dfrac{b_x}{m_{eq}} & \dfrac{K_m}{m_{eq} r_m} & 0 & 0 & 0 \\ 0 & -\dfrac{2K_m}{r_m L_m} & -\dfrac{R_m}{L_m} & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & -\dfrac{b_y}{m_{eq}} & \dfrac{K_m}{m_{eq} r_m} \\ 0 & 0 & 0 & 0 & -\dfrac{2K_m}{r_m L_m} & -\dfrac{R_m}{L_m} \end{array}\right], \qquad \tilde{\mathbf{B}} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ \dfrac{1}{L_m} & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & \dfrac{1}{L_m} \end{bmatrix}$$

1.3 Drawing shapes

Rearranging the decoupled equations of motion (1.21), we can express the currents needed to follow a prescribed motion of the tool head:

$$\boxed{\begin{aligned} i_x &= \frac{m_{eq} r_m}{K_m}\,\ddot{x} + \frac{b_x r_m}{K_m}\,\dot{x} \\ i_y &= \frac{m_{eq} r_m}{K_m}\,\ddot{y} + \frac{b_y r_m}{K_m}\,\dot{y} \end{aligned}} \tag{1.23}$$

The motor equations for each motor are:

$$\begin{aligned} u_{m,L} &= L_m \frac{di_L}{dt} + R_m i_L + K_m \dot{\theta}_L = L_m \frac{di_L}{dt} + R_m i_L + \frac{K_m}{r_m}\,(\dot{x} + \dot{y}) \\ u_{m,R} &= L_m \frac{di_R}{dt} + R_m i_R + \frac{K_m}{r_m}\,(\dot{x} - \dot{y}) \end{aligned}$$

Adding them, and using (1.19):

$$u_{m,x} = u_{m,L} + u_{m,R} = L_m \left[\frac{di_L}{dt} + \frac{di_R}{dt}\right] + R_m\,[i_L + i_R] + \frac{2K_m}{r_m}\,\dot{x}$$

and similarly by subtracting them, we get the decoupled input voltages, consistent with (1.20):

$$\boxed{\begin{aligned} u_{m,x} &= L_m \frac{di_x}{dt} + R_m i_x + \frac{2K_m}{r_m}\dot{x} \\ u_{m,y} &= L_m \frac{di_y}{dt} + R_m i_y + \frac{2K_m}{r_m}\dot{y} \end{aligned}} \tag{1.24}$$

The original motor voltages then follow from (1.19):

$$u_{m,L} = \tfrac{1}{2}\left(u_{m,x} + u_{m,y}\right), \qquad u_{m,R} = \tfrac{1}{2}\left(u_{m,x} - u_{m,y}\right) \tag{1.25}$$

Since (1.24) contains $\frac{di_x}{dt}$, and (1.23) contains $\ddot{x}$, the voltages depend on the third derivative of the position. Therefore, for any C^3 functions $x(t)$ and $y(t)$ describing the position of the tool head, we can derive an open-loop analytical solution for the input voltages, even if those voltages aren't realistic.

Ellipse

An ellipse with centre (h, k) and semi-axes a and b has the parametric form:

$$x(t) = h + a\cos t, \qquad y(t) = k + b\sin t \tag{1.26}$$

We also need the derivatives up to the third order:

$$\begin{aligned} \dot{x} &= -a\sin t, \quad \ddot{x} = -a\cos t, \quad \dddot{x} = a\sin t \\ \dot{y} &= b\cos t, \quad \ddot{y} = -b\sin t, \quad \dddot{y} = -b\cos t \end{aligned}$$

Substituting into (1.23):

$$\begin{aligned} i_x(t) &= -\frac{m_{eq} r_m}{K_m} a\cos t - \frac{b_x r_m}{K_m} a\sin t = -\frac{r_m}{K_m} a\left[m_{eq}\cos t + b_x \sin t\right] \\ i_y(t) &= \frac{r_m}{K_m} b\left[b_y \cos t - m_{eq}\sin t\right] \end{aligned}$$

And into (1.24):

$$\begin{aligned} u_{m,x}(t) &= L_m \frac{r_m}{K_m} a\left[m_{eq}\sin t - b_x\cos t\right] - R_m \frac{r_m}{K_m} a\left[m_{eq}\cos t + b_x \sin t\right] - \frac{2K_m}{r_m} a\sin t \\ u_{m,y}(t) &= -L_m \frac{r_m}{K_m} b\left[m_{eq}\cos t + b_y \sin t\right] + R_m \frac{r_m}{K_m} b\left[b_y\cos t - m_{eq}\sin t\right] + \frac{2K_m}{r_m} b\cos t \end{aligned}$$

With

$$u_{m,L}(t) = \tfrac{1}{2}\left(u_{m,x}(t) + u_{m,y}(t)\right), \qquad u_{m,R}(t) = \tfrac{1}{2}\left(u_{m,x}(t) - u_{m,y}(t)\right)$$

we obtain the analytical open-loop solution for drawing the ellipse.

2 Finite State Machine

3 Hybrid System

References